

Introduction to Computer Programming (ICP, CCIT4020)

HKU SPACE Community College, 2021-2022 Semester 1

(Lab 04)

(Practicing Lab, NOT for submission)

Review the related lecture note(s), and finish exercises below.

1. Write a Python program to demonstrate the use of **if-elif-else** structure to **input and compare** two floating numbers **x** and **y**. (named **Lab4_1.py**):

- **if** **x** and **y** are equal, display two numbers and show they are equal,
- **elif** number **x** is greater than number **y**, display a message with two numbers
- **else** number **x** is less than number **y**, display a message with two numbers

(Remark: Display the floating point numbers with 1 decimal place.)

Sample Program Output

```
Comparing two numbers:
Enter 1st Float number: 9
Enter 2nd Float number: 9
The entered numbers are 9.0 and 9.0
9.0 equals 9.0
```

←--- User Input, with <ENTER>

```
Comparing two numbers:
Enter 1st Float number: 2.3
Enter 2nd Float number: 6.8
The entered numbers are 2.3 and 6.8
2.3 is less than 6.8
```

```
Comparing two numbers:
Enter 1st Float number: 4.5
Enter 2nd Float number: 2.3
The entered numbers are 4.5 and 2.3
4.5 is greater than 2.3
```

2. Write a Python program (named **Lab4_2.py**):
- Ask user 2 questions, and let the user input information about them
 - **What is your gender, m/f?**
 - **How old are you?**
 - Based on the user input, show the corresponding response message (See below table)
 - Use **if-elif-else** structure with relational and logical operators to finish this task

Input One (Gender)	Input Two (Age) (in integer)	Response Message
m	18 or below	"That's Great!"
m	Above 18	"Working fine."
f	18 or below	"Keep Enjoying!"
f	Above 18	"You're doing fine."
All other unexpected Situations		"Have a nice day!"

Sample program output

```
What is your gender, m/f? m
How old are you? 9
That's Great!
```

← User Input, with <ENTER>

```
What is your gender, m/f? f
How old are you? 19
You're doing fine.
```

3. A student's letter grade is calculated according to the following table.

Write a Python program (named `Lab4_3.py`) that accepts a student's numerical grade (in floating point number), converts and assigns the numerical grade to an equivalent letter grade using *nest if* structure, and displays the letter grade accordingly.

Numerical Grade	Letter Grade
From 100.0, to greater than or equal to 90.0	A
less than 90.0, but greater than or equal to 80.0	B
less than 80.0, but greater than or equal to 70.0	C
less than 70.0, but greater than or equal to 60.0	D
less than 60.0, but greater than or equal to 50.0	E
less than 50.0, but greater than or equal to 0.0	F
Out of normal range (thus, >100 or <0)	U

Please try different grades in your testing (see sample output below). E.g. 91.5, 50.0 and 49.9

Sample program output

```
Enter the student's numerical grade: 90.5
The student receives a grade of A
```

← User Input, with <ENTER>

```
Enter the student's numerical grade: 50
The student receives a grade of E
```

```
Enter the student's numerical grade: 49.9
The student receives a grade of F
```

```
Enter the student's numerical grade: 102
The student receives a grade of U
```